Batterjee Medical College
For Science and Technology

BMC

Pneumonia

An approach to a Case of Cough

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BATTERJEE MEDICAL COLLEGE

Learning Objectives

Knowledge:

- Define pneumonia and describe its underlying pathophysiology involving alveolar and interstitial tissue inflammation.
- Classify pneumonia based on mode of acquisition, causative microbe, and anatomical site.
- Identify the common etiologic agents responsible for community-acquired pneumonia (typical vs. atypical) and hospital-acquired pneumonia.
- Differentiate between typical and atypical CAP in terms of onset, symptoms, Gram stain findings, diagnostic imaging, and penicillin responsiveness.
- List the predisposing risk factors for pneumonia and pulmonary aspiration.
- Recognize potential complications of pneumonia.
- Describe prevention strategies, including indication guidelines for the pneumococcal vaccine, influenza vaccination, and smoking cessation.

Skills:

- Interpret clinical manifestations and objective findings to evaluate cases of cough and suspected pneumonia.
- Analyze diagnostic imaging, including identifying lobar consolidation, air bronchograms, and reticulonodular infiltrates on chest X-rays and CT scans.
- Calculate and apply clinical risk scores (such as CURB-65) to determine appropriate disposition (outpatient, inpatient, or ICU).
- Formulate an evidence-based empiric management plan tailored to patient risk factors, disease severity, and specific causative pathogens.
- Assess multidrug-resistant (MDR) pathogen risk factors to select appropriate IV dual/triple empiric antibiotic regimens for **hospital-acquired pneumonia.**

Value:

- Demonstrate a professional attitude by strictly adhering to training facility rules and regulations during clinical education.

Main Subtopics

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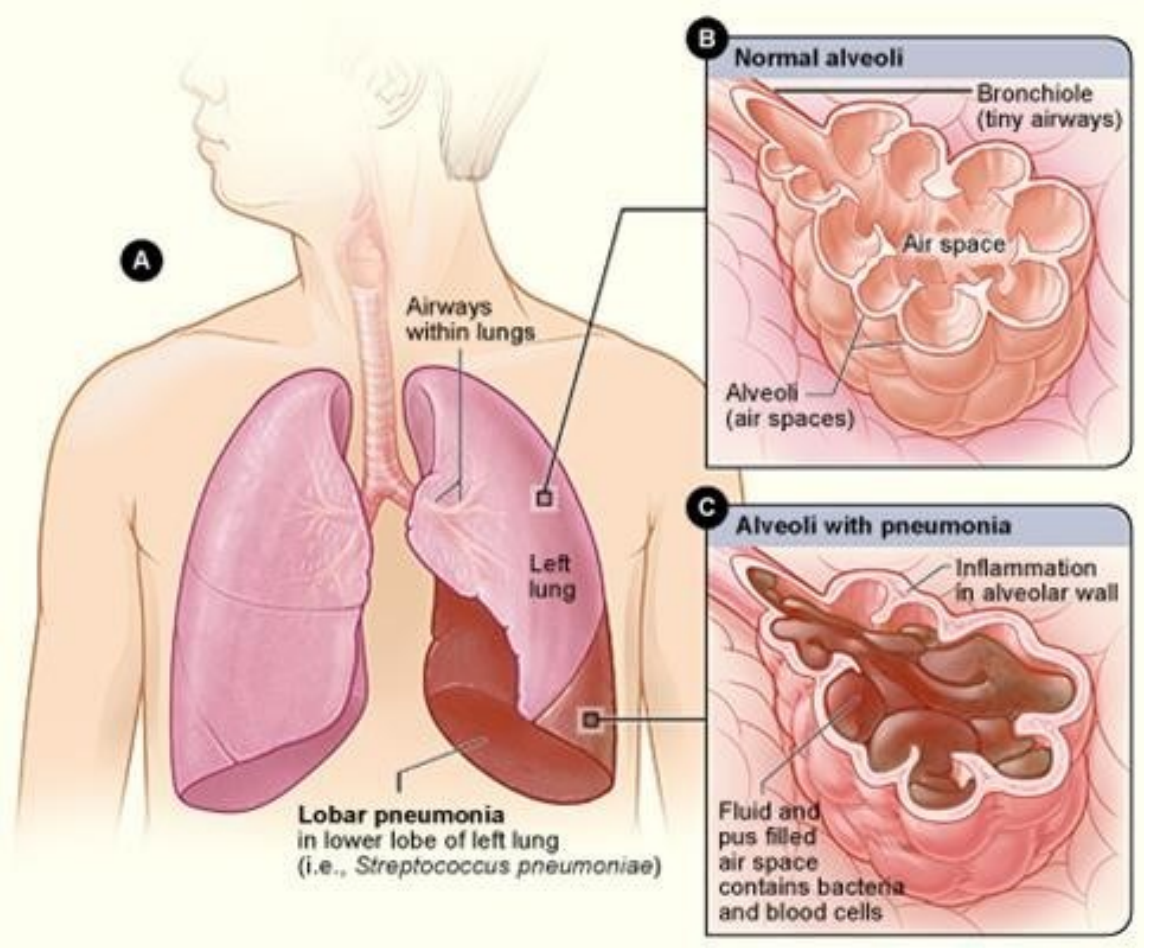
- Community acquired pneumonia (definition, Types, pathogenesis, clinical presentation, diagnosis, management)
- Hospital acquired pneumonia (definition, Types, pathogenesis, clinical presentation, diagnosis, management)
- Atypical pneumonia (definition, Types, pathogenesis, clinical presentation, diagnosis, management)
- Aspiration pneumonia (definition, Types, pathogenesis, clinical presentation, diagnosis, management)

Pneumonia

Definition:

A respiratory infection characterized by inflammation of the alveolar space and/or the interstitial tissue of the lungs

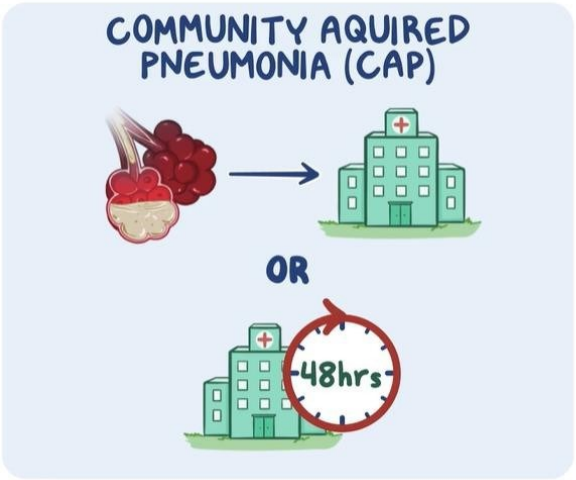
It is most commonly transmitted via aspiration of airborne pathogens (primarily bacteria, but also viruses and fungi) but may also result from the aspiration of stomach contents.

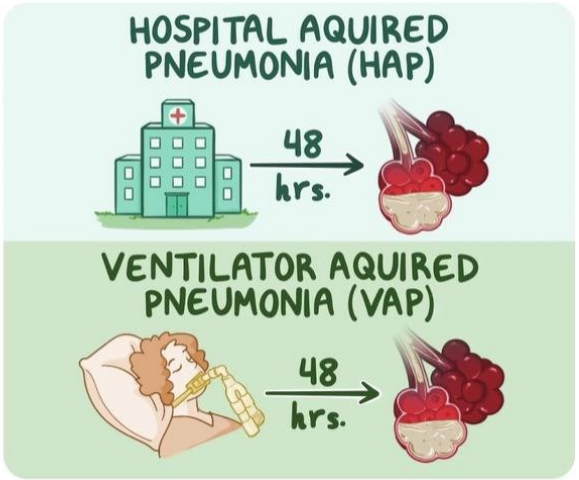


Pneumonia

Classification:

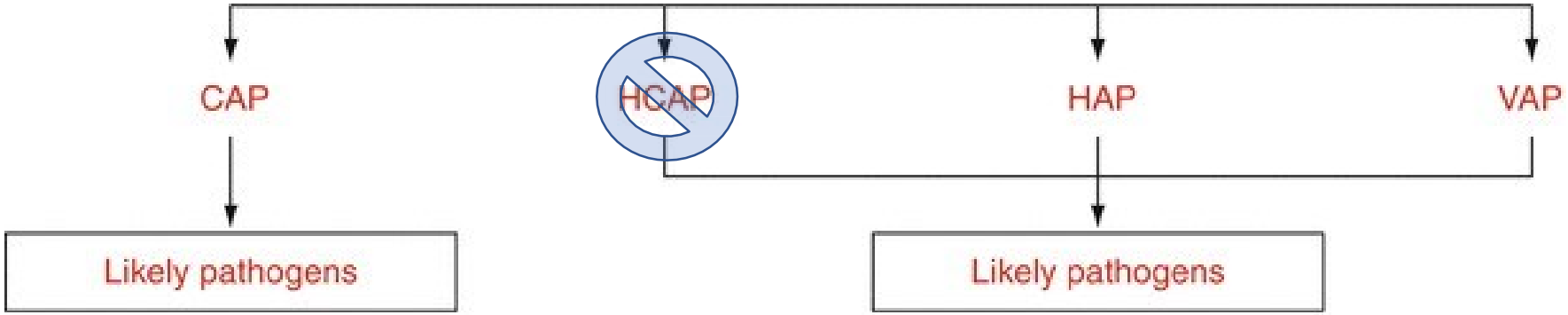
- Acquisition: CAP Vs. HAP
- Microbe: Bacterial Vs. Viral Vs. Fungal
- Location: Bronchopneumonia Vs. Lobar Vs. interstitial (Atypical)
- Aspiration





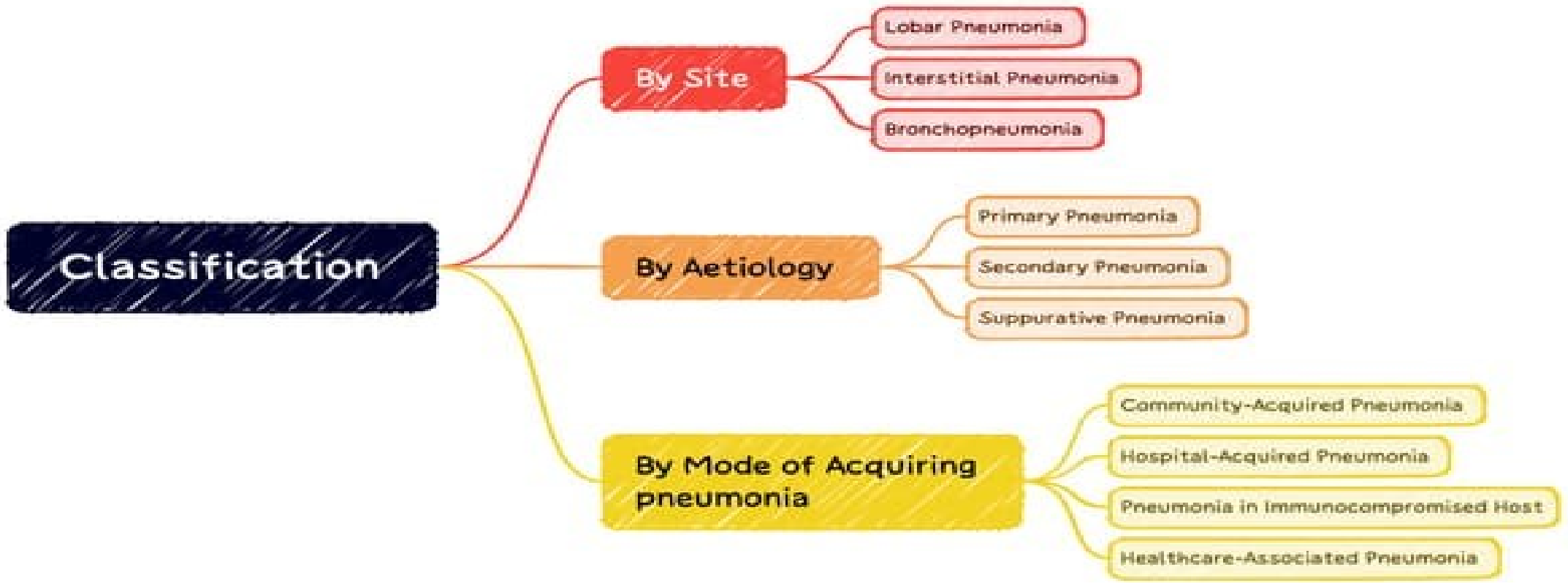
Pneumonia

Etiology:

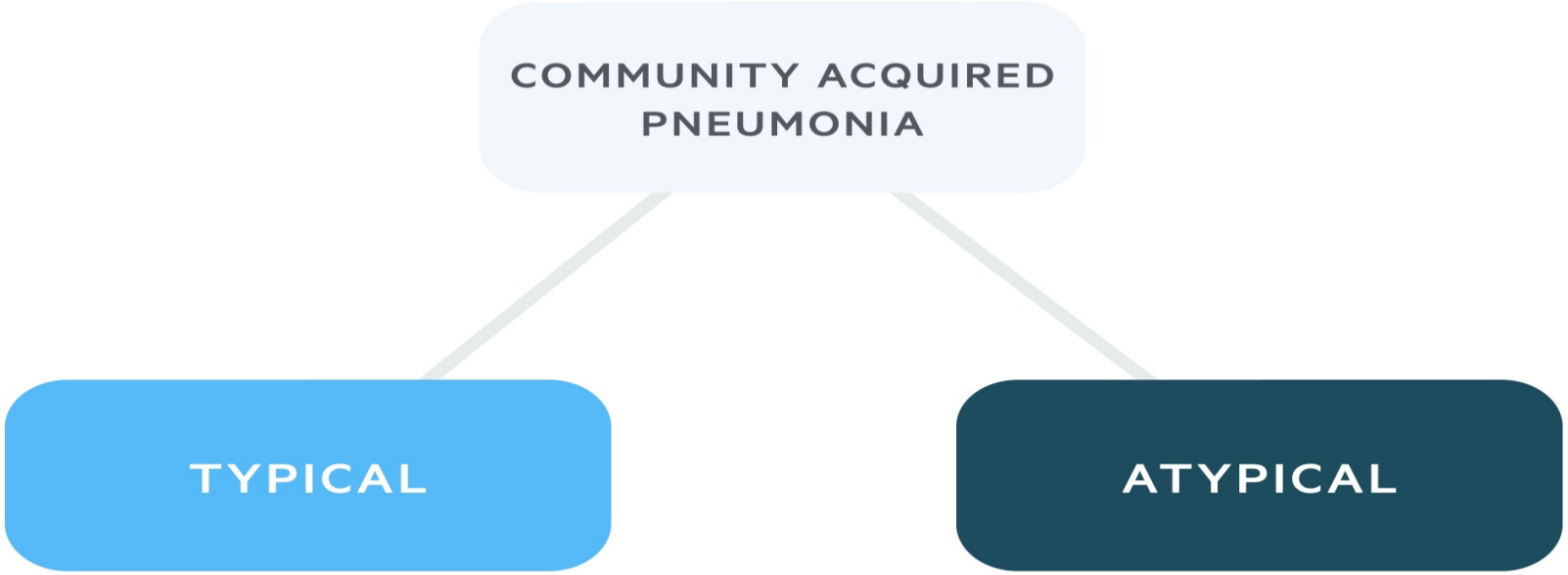


- *Streptococcus pneumoniae*
- *Hemophilus influenzae*
- Methicillin-susceptible *Staphylococcus aureus*
- *Mycoplasma pneumoniae*
- *Chlamydia pneumonia*
- *Legionella* species
- Respiratory virus (influenza, RSV)
- Community-acquired methicillin-resistant *Staphylococcus aureus* (cMRSA)
- Antibiotic-susceptible *Enterobacteriaceae*

- *Pseudomonas aeruginosa*
- Methicillin-resistant *Staphylococcus aureus*
- *Acinetobacter* species
- Streptococcus pneumoniae
- Antibiotic-resistant *Enterobacteriaceae*


Classification
By Site
Lobar Pneumonia
Interstitial Pneumonia
Bronchopneumonia
By Aetiology
Primary Pneumonia
Secondary Pneumonia
Suppurative Pneumonia
By Mode of Acquiring pneumonia
Community-Acquired Pneumonia
Hospital-Acquired Pneumonia
Pneumonia in Immunocompromised Host
Healthcare-Associated Pneumonia
Presented with xmind

Community acquired



Typical CAP

Common agents:

- *S. pneumoniae* (60%)
- *Haemophilus influenzae* (15%)
- Aerobic gram-negative rods (6% to 10%)—*Klebsiella* (and other Enterobacteriaceae)
- *S. aureus* (2% to 10%)

What is the difference between typical and atypical community-acquired pneumonia?

Variable	Typical	Atypical
Etiology	S.pneumoniae, H.influenza	Mycoplasma pneumoniae, chlamydophila pneumoniae , legionella, TB, viral or fungal
Clinical presentation	Sudden onset of fever, chill, productive cough, shortness of breath and chest pain	Gradual onset headache, sore throat and body ache
Diagnosis Gram Stain	Useful	Useless (no cell wall)
Radiography	Lobar infiltrate	Dramatic changes: patchy or interstitial
Treatment with penicillin	Sensitive	Resistant

Pneumonia

Risk factors

- **Old age and immobility of any cause**
- **Chronic diseases**
  - Preexisting cardiopulmonary conditions (e.g., bronchial asthma, COPD, heart failure)
  - Acquired or congenital abnormalities of the air ways (e.g., bronchiectasis, space-occupying lesions, cystic fibrosis)
- **Immunosuppression**
  - HIV infection, Diabetes mellitus, immunosuppressive therapy
- **Impaired airway protection**
  - Alteration in consciousness (e.g., due to stroke, seizure, anesthesia, drugs, alcohol)
- **Environmental factors**
  - Crowded living conditions & Contaminated water systems (e.g., in hotels, on cruise ships)
- **Surgical procedures**



Typical CAP

Symptoms:

- Acute onset of fever and shaking chills
- Cough productive of thick, purulent sputum
- Pleuritic chest pain
- Dyspnea

Signs:

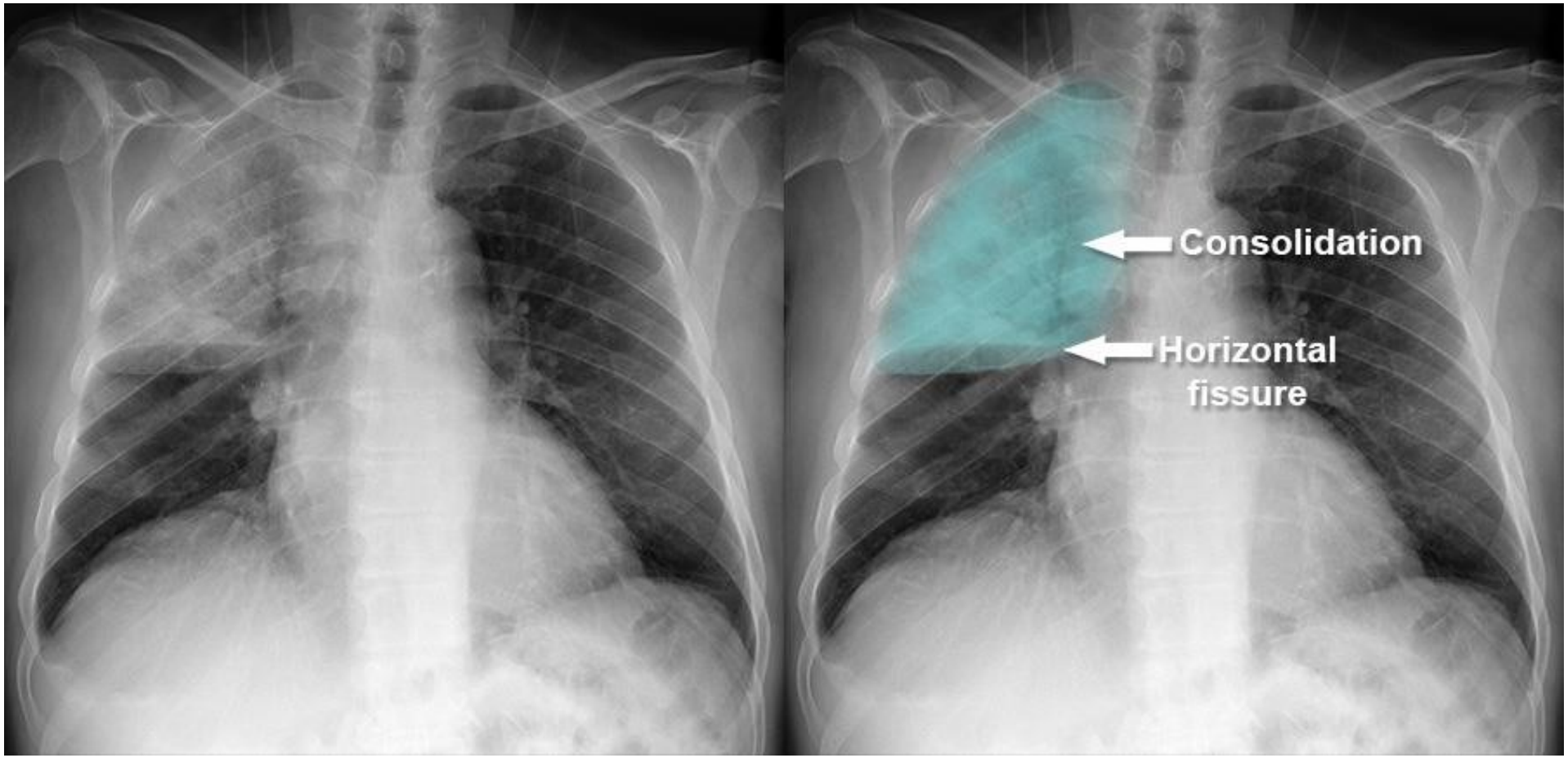
- Tachycardia, tachypnea
- Late inspiratory crackles, bronchial breath sounds, increased tactile and vocal fremitus, dullness on percussion
- Pleural friction rub (associated with pleurisy)

Chest radiograph (CXR)

- Lobar consolidation
- Multilobar consolidation indicates very serious illness

Typical CAP

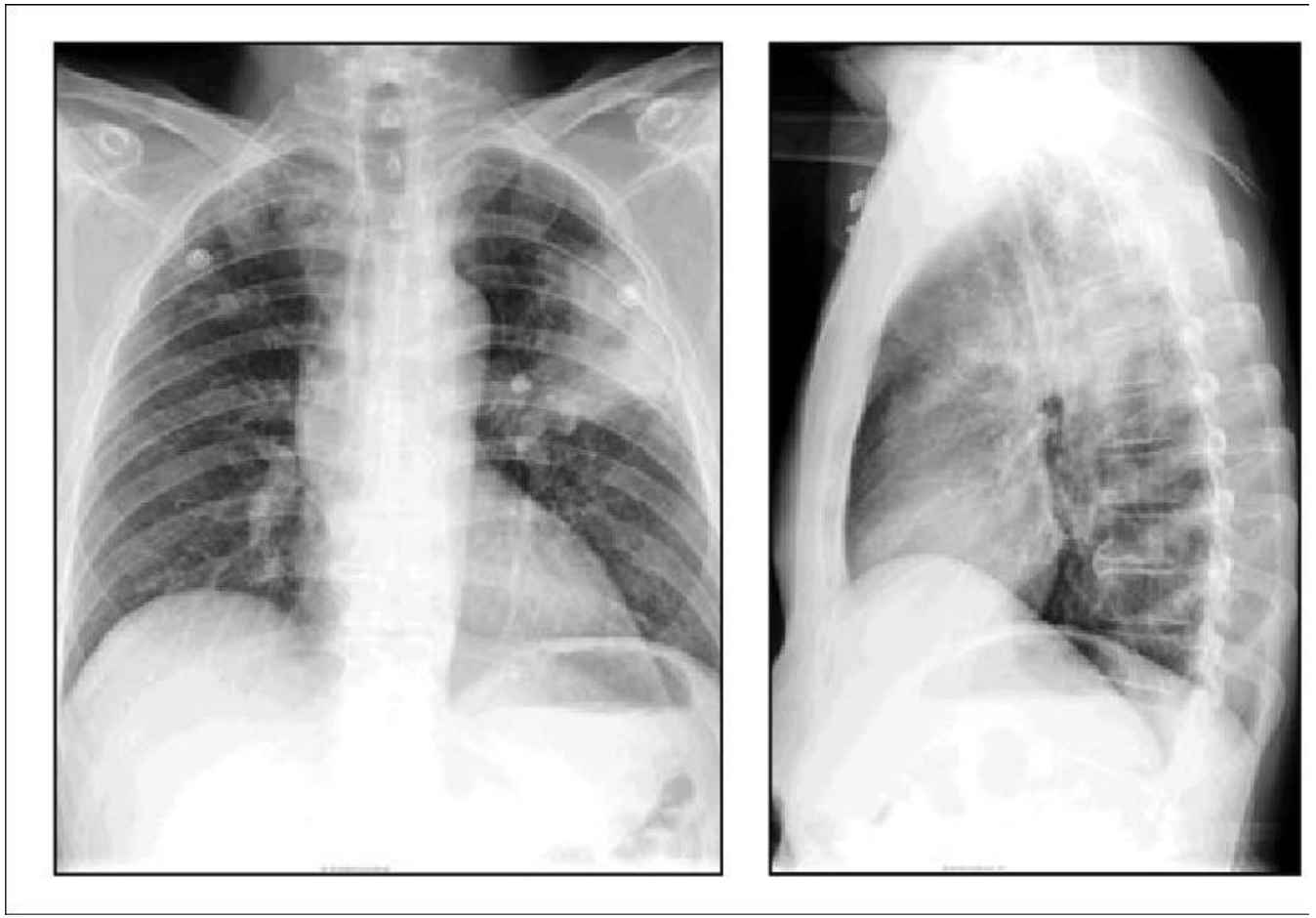
Lobar PNA:



Typical CAP

Lobar PNA:





Atypical CAP “Walking Pneumonia”

Common agents:

- *Mycoplasma pneumoniae* (most common)
- *Chlamydia pneumoniae*
- *Chlamydia psittaci*
- *Coxiella burnetii* (Q fever)
- *Legionella* spp.
- Viruses: influenza virus (A and B), adenoviruses, parainfluenza virus, RSV

Quick HIT

“Atypical” pneumonia refers to organisms not visible on Gram stain and not culturable on standard blood agar.

Atypical CAP “Walking Pneumonia”

Symptoms:

- Insidious onset—headache, sore throat, fatigue, myalgias
- Dry cough (no sputum production)
- Fevers (chills are uncommon)

Signs:

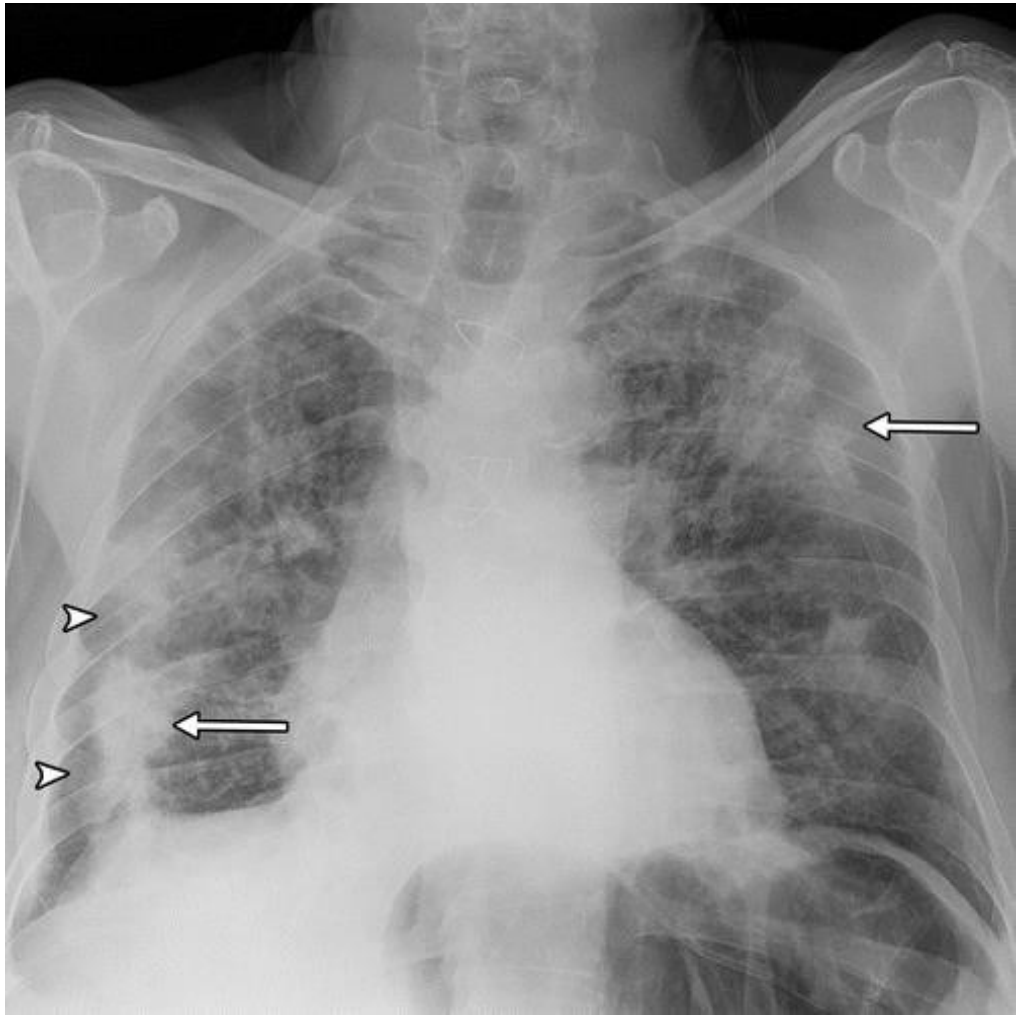
- Pulse-temperature dissociation—normal pulse in the setting of high fever is suggestive of atypical CAP
- Wheezing, rhonchi, crackles

Chest radiograph (CXR)

- Diffuse reticulonodular infiltrates
- Absent or minimal consolidation

Atypical CAP





Atypical CAP

"ATYPICAL"/"WALKING" PNEUMONIA

- MYCOPLASMA PNEUMONIAE



~ INSIDIOUS ONSET
~ HEADACHE
~ NONPRODUCTIVE COUGH
~ NO FEVER

BACTERIA without a CELL WALL

OCCURS in AREAS w/ MANY PEOPLE PACKED TOGETHER
- MILITARY CAMPS
- PRISONS





LEGIONELLA PNEUMOPHILA



~ HEADACHES
~ MILD COUGH
~ CONFUSION
~ HIGH FEVER
~ WATERY DIARRHEA



- WATER SYSTEMS
- WATER-BASED COOLING SYSTEMS in HOTELS



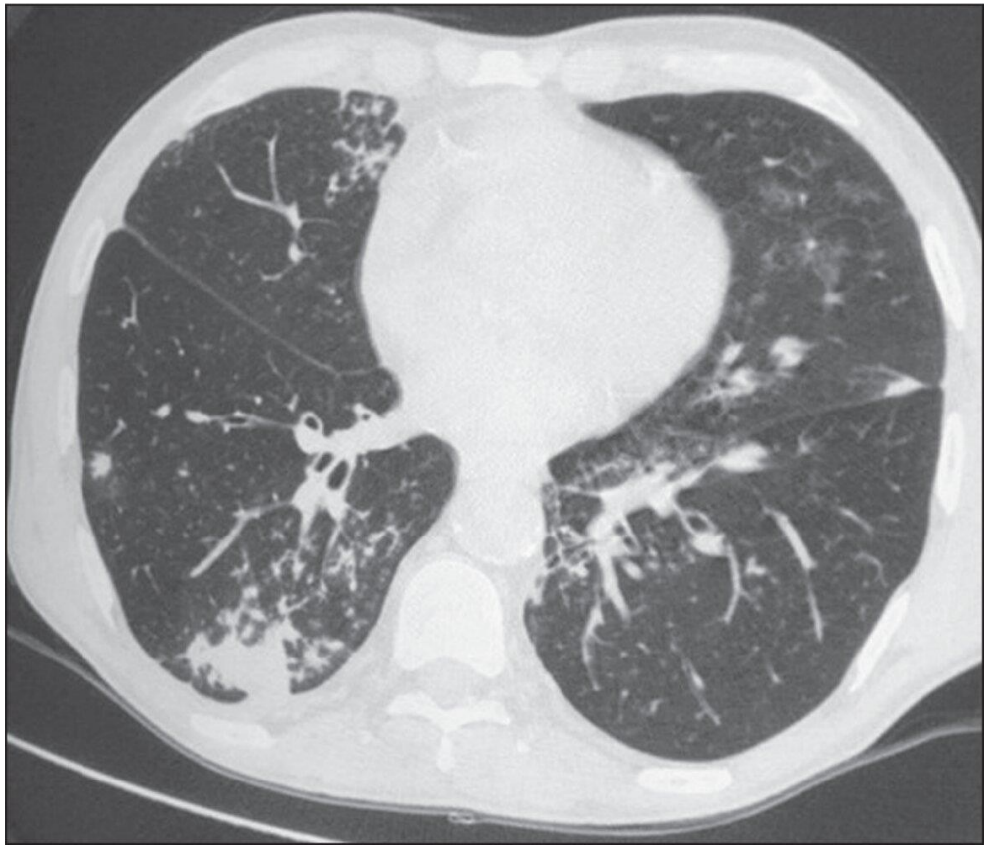
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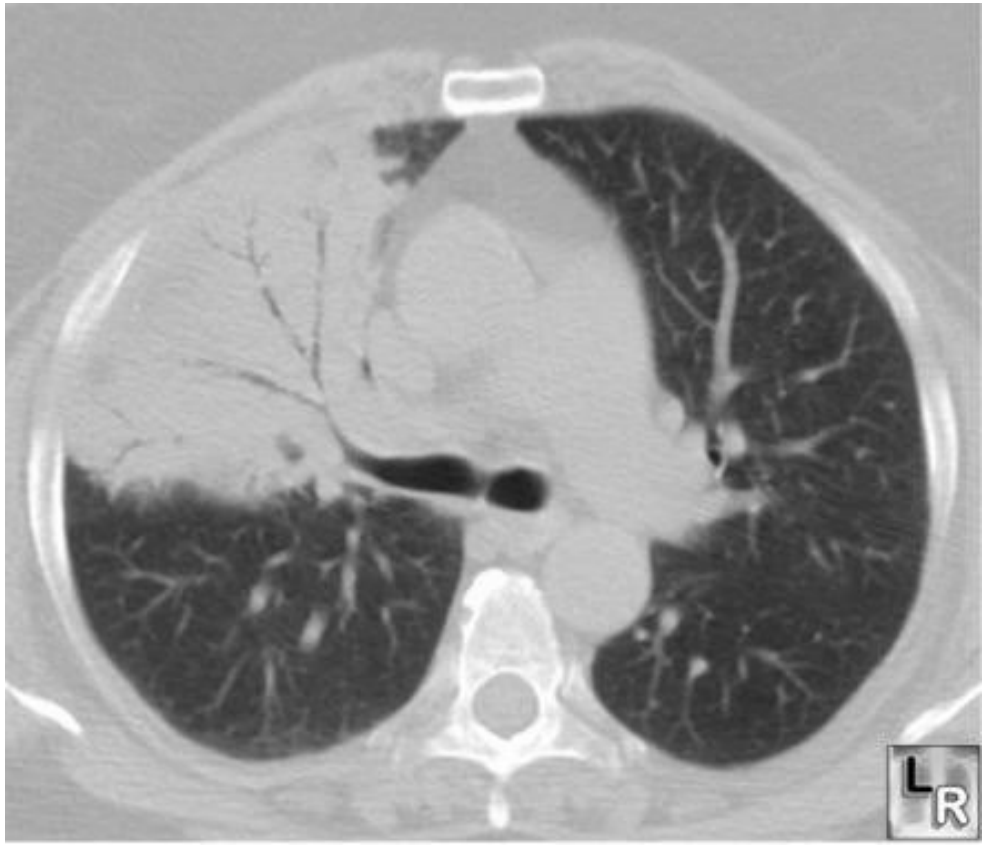
Diagnostics:

- CXR (PA & lateral)
- CBC
- ↑ CRP, ↑ ESR
- ↑ Serum procalcitonin
- Chest CT
- Sputum gram stain & cultures
- Flu Swab
- Blood cultures, sputum culture
- Influenza and COVID-19 testing
- ↓ SpO_2
- Urinary antigen assay for *Legionella (When suspected)*

Pneumonia

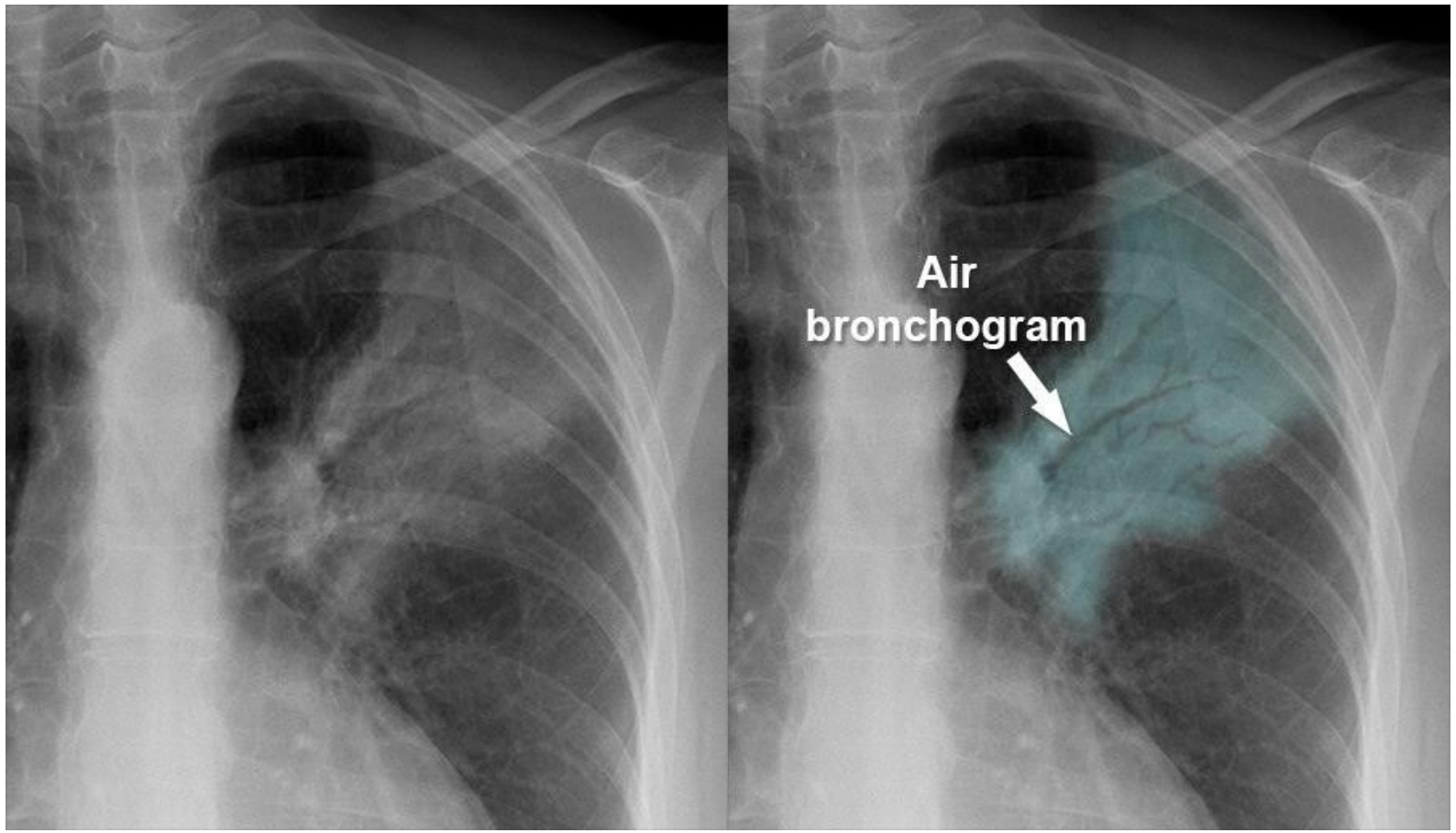
CT Chest (w/o contreast)





Pneumonia

Air Bronchogram:



Pneumonia

Management:

Inpatient Vs. Out

Table 1. CURB-65 Scoring[46]

Symptom	Points
Confusion	1
Urea: BUN >19 mg/dL (>7 mmol/L)	1
Respiratory rate ≥30 breaths/min	1
Systolic BP <90 mm Hg or diastolic BP ≤60 mm Hg	1
Age ≥65 years	1
Total	_____

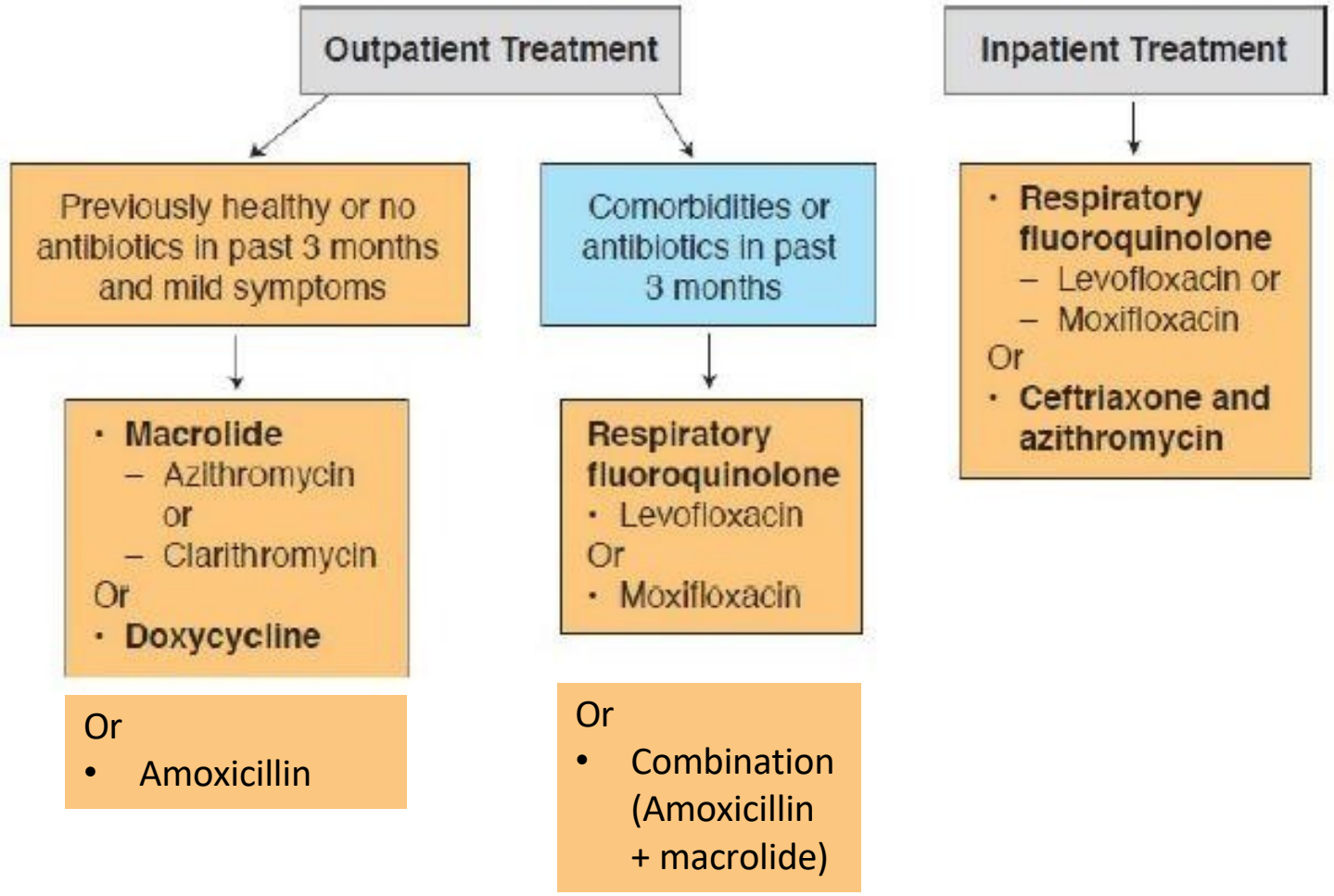
Score	Risk	Disposition
0 or 1	1.5% mortality	Outpatient care
2	9.2% mortality	Hospitalization is indicated.
≥3	22% mortality	Consider ICU admission.

Abbreviations: BP, blood pressure; BUN, blood urea nitrogen; ICU, intensive care unit.

Pneumonia

Empiric Management:

- Begin empiric antibiotic therapy based on severity and patient risk factors (e.g., VAP vs. CAP).
- Consider antiviral therapy for influenza and COVID-19 as indicated.
- Identification and management of sepsis
- Fluid resuscitation and management of septic shock
- Respiratory support if needed
- Supplemental oxygen for hypoxia



Pneumonia

CAP Management:

TABLE 94.1 Treatment of Pneumonia by Specific Etiologic Agent

Etiologic Agent	Preferred Antimicrobial	Alternative Antimicrobial
Streptococcus pneumoniae	Penicillin	Cephalosporin, moxifloxacin, levofloxacin
Haemophilus influenzae	Cefuroxime, ceftriaxone	
Mycoplasma pneumoniae	Macrolide	Moxifloxacin, levofloxacin
Legionella	Macrolide or quinolone	
Staphylococcus aureus		
Methicillin-susceptible	Nafcillin	Cephalosporin
Methicillin-resistant	Vancomycin (IV)	Doxycycline or TMP/SMX (oral)
Moraxella catarrhalis	Amoxicillin/clavulanate, cefuroxime, ceftriaxone, TMP/SMX	

TMP/SMX, Trimethoprim/sulfamethoxazole

Primary and Secondary Prevention

Pneumococcal Vaccine:

Given to all patients with CAP prior to discharge unless already vaccinated.

Influenza Vaccine

Smoking Cessation

Indications for pneumococcal vaccination in adults in the United States

All adults ≥65 years of age

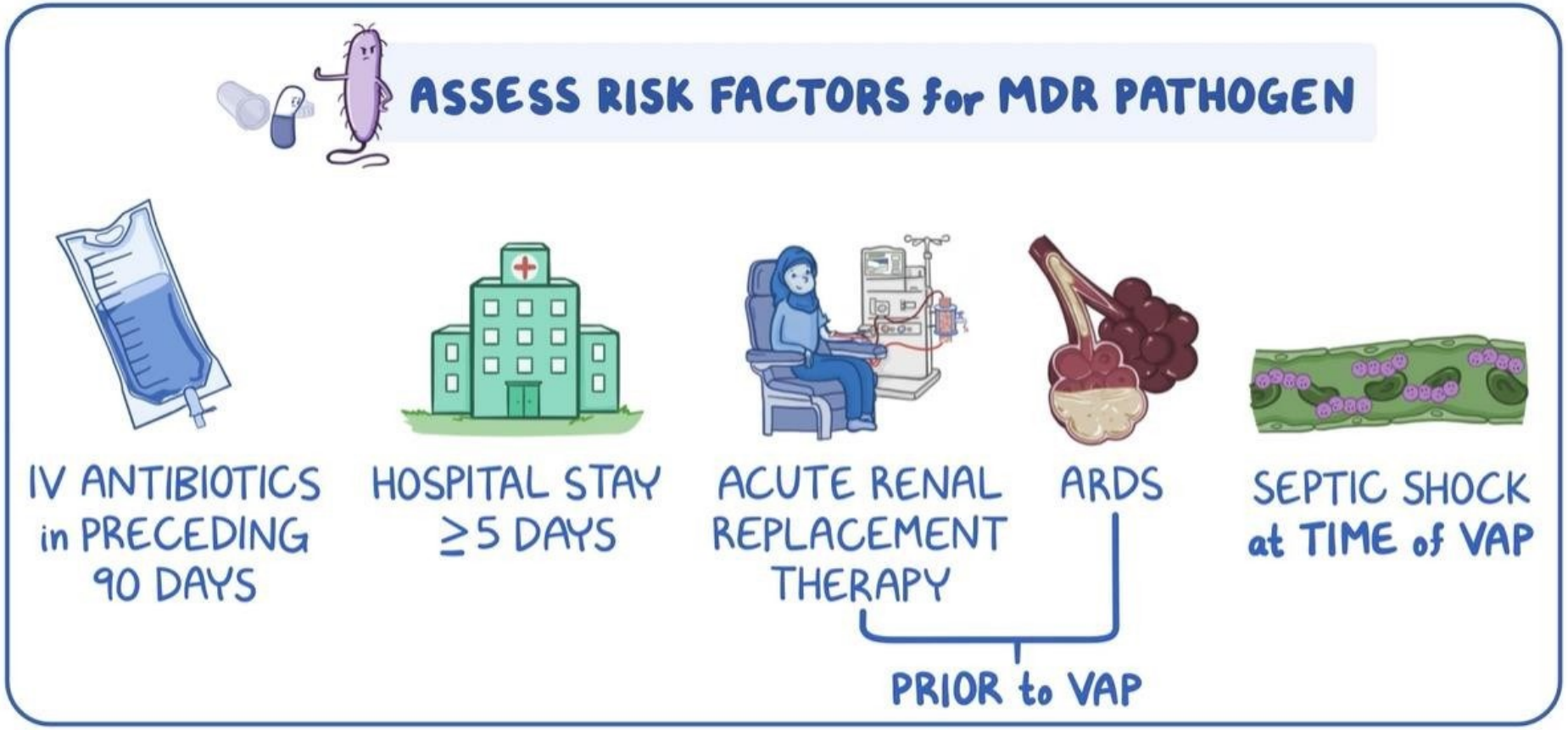
Adults 19 to 64 years of age with any of the following:

- Predisposing medical conditions:
  - Alcohol use disorder
  - Chronic heart disease*
  - Chronic lung disease¶
  - Chronic liver disease
  - Diabetes mellitus
  - Sickle cell disease or other hemoglobinopathies
  - Current cigarette smoking
- Increased risk of meningitis:
  - Cerebrospinal fluid leak
  - Cochlear implant
- Immunocompromising conditions and other conditions associated with altered immunocompetence△:
  - Congenital or acquired immunodeficiency◇
  - Generalized active malignancy
  - Human immunodeficiency virus infection§
  - Iatrogenic immunosuppression¥
  - Hodgkin disease
  - Leukemia
  - Lymphoma
  - Multiple myeloma
  - Solid organ transplant
  - Chronic kidney disease‡ and nephrotic syndrome
  - Functional or anatomic asplenia
- History of invasive pneumococcal disease†

Pneumonia

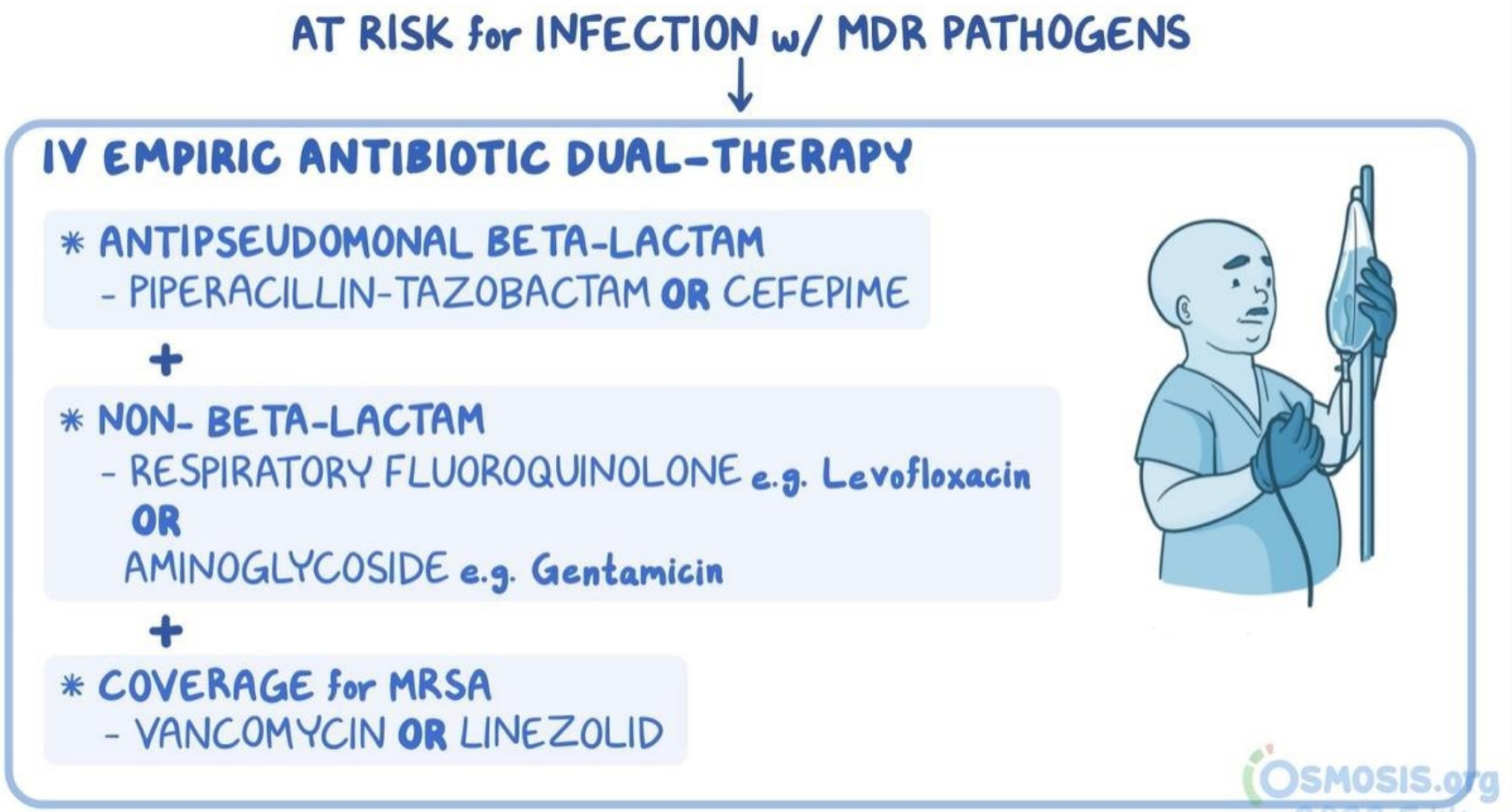
Hospital-acquired pneumonia (HAP): pneumonia onset > 48 hours after a patient is admitted to a hospital without evidence of disease at the time of admission

HAP Management:



Pneumonia

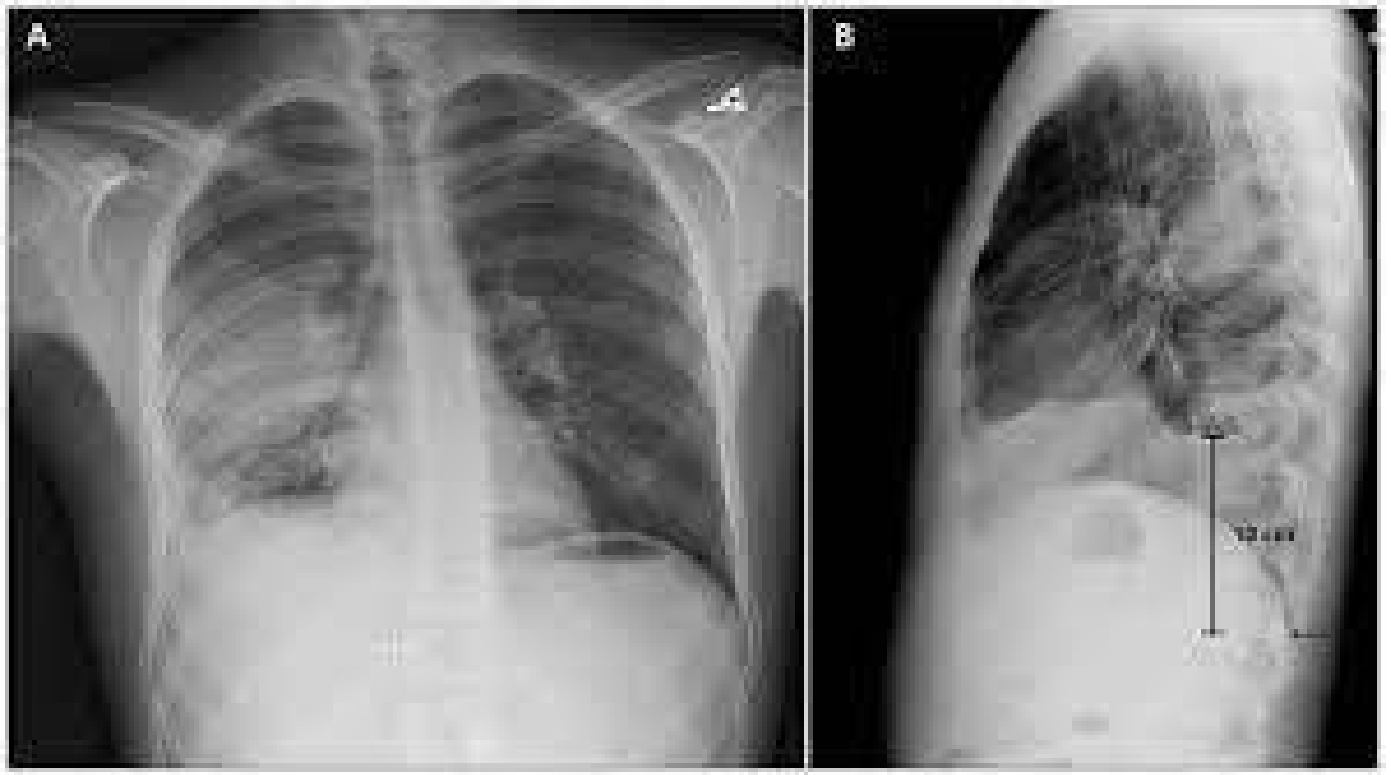
HAP Management:



Pneumonia Complications

Complications:

- Parapneumonic pleuritis
- Parapneumonic pleural effusion
- Pleural empyema
- Lung abscess
- ARDS
- Respiratory failure
- Sepsis



Summary

Respiratory

Pneumonia

lung parenchyma inflammation due to lower respiratory tract infection, leading to puss-filled alveoli and consolidation.

- Can result in reduced perfusion (hypoxia), as well as restricted ventilation from pain / effusion (T2 RF).
- Can lead to effusion, lung abcess, and Pericarditis.

° Types:

Community-Acquired Pneumonia (CAP): develops outside of hospital.
- most commonly caused by Strep, followed by viral.

Hospital-Acquired Pneumonia (HAP): develops more than 48h after hospital admission.
- Ventilator-associated Pneumonia - develops after intubation.
- Guidelines based on VAP causes, which are completely different to CAP.

° Diagnosis: • Investigations: - CXR - Sputum Culture - SpO_2, ABG - FBC, CRP, LFT, U+E

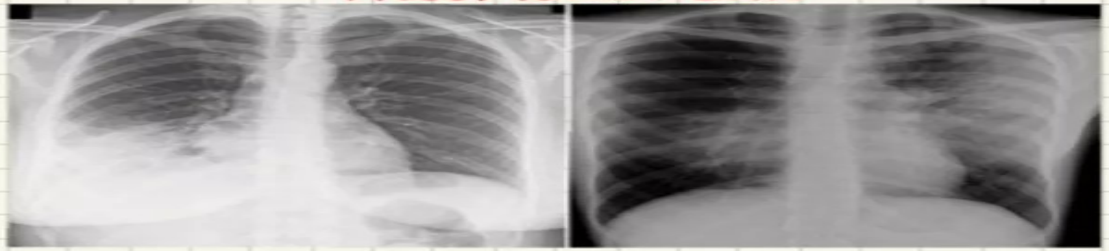
CAP:
1. Lower respiratory tract infection symptoms: cough, sputum, Pleuritic P, SOB.
2. Systemic symptom: fever, Temp >38°C, chills, tachycardia.
3. New chest exam signs: reduced expansion, dull percussion, crackles.
4. No other likely cause.

HAP:
Lower respiratory tract infection symptoms
+ CXR Opacity without other explanation (e.g. oedema, fibrosis, etc.)

° Management:

- O_2, Fluids, Analgesia.
- IV Antibiotics
  → CAP: Amoxicillin
  → HAP: based on trust guidelines, adjust on culture results.
- Manage complications e.g. drain effusion / abcess.

PNEUMONIA CXR



May take 3-6 months for all symptoms to resolve.

Case Scenario 1

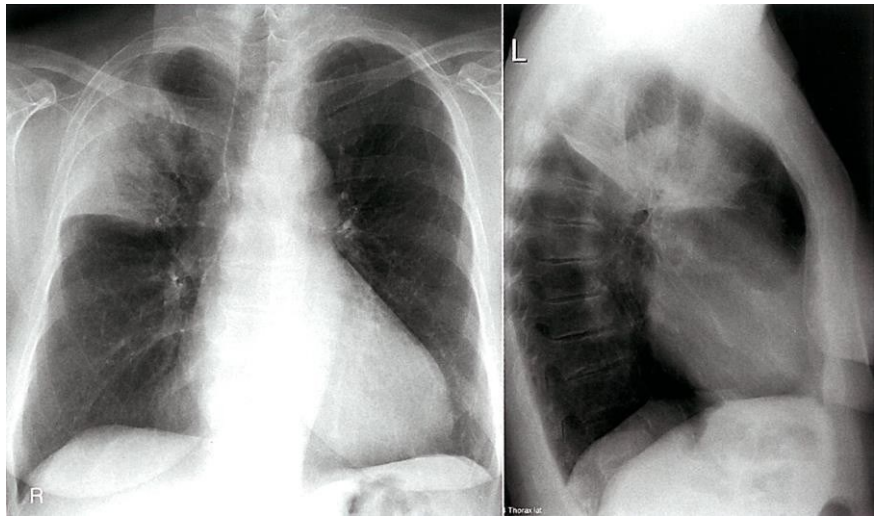
A 53-year-old man comes to the physician because of a 1-day history of fever and chills, severe malaise, and cough with yellow-green sputum. He works as a commercial fisherman on Lake Superior. Current medications include metoprolol and warfarin. His temperature is 38.5°C (101.3°F), pulse is 96/min, respirations are 26/min, and blood pressure is 98/62 mm Hg. Examination shows increased fremitus and bronchial breath sounds over the right middle lung field. An x-ray of the chest shows consolidation of the right upper lobe. Which of the following is the most likely causal pathogen?

a. *Mycobacterium tuberculosis*
b. *Pseudomonas aeruginosa*
c. *Fusobacterium nucleatum*
d. *Mycoplasma pneumoniae*
e. *Legionella pneumophila*
f. *Staphylococcus aureus*
g. *Streptococcus pyogenes*
h. *Haemophilus influenzae*
i. *Histoplasma capsulatum*
j. *Streptococcus pneumoniae*

Case Scenario 2

A 67-year-old woman is brought to the emergency department for the evaluation of fever, chest pain, and a cough productive of a moderate amount of greenish-yellow sputum for 2 days. During this period, she has had severe malaise, chills, and difficulty breathing. She has hypertension, hypercholesterolemia, and type 2 diabetes mellitus. She smoked one pack of cigarettes daily for 20 years, but quit 5 years ago. Current medications include simvastatin, captopril, and metformin. Temperature is 39°C (102.2°F), pulse is 110/min, respirations are 33/min, and blood pressure is 143/88 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. Crackles are heard on auscultation over the right upper lung field. Laboratory studies show a leukocyte count of 12,300/mm^3, an erythrocyte sedimentation rate of 60 mm/h, and a urea nitrogen of 15 mg/dL. A chest x-ray is shown. Which of the following is the most appropriate next step in the management?

a. Outpatient treatment with moxifloxacin
b. Inpatient treatment with cefepime, azithromycin, and gentamicin
c. ICU admission and administration of ceftriaxone and azithromycin
d. Outpatient treatment with azithromycin and amoxicillin/clavulanate
e. Inpatient treatment with azithromycin and cefotaxime
f. ICU admission and administration of ampicillin/sulbactam and levofloxacin
g. Outpatient treatment with azithromycin
h. Inpatient treatment with ceftriaxone
i. ICU admission and administration of aztreonam and levofloxacin



Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine
  20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board
  4th edition
- Osmosis

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Thank You